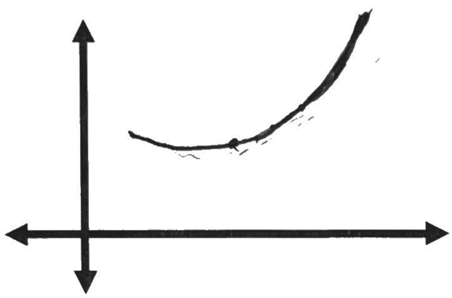
The Second Derivative Test for Concavity

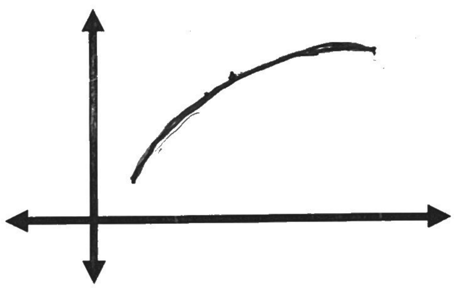
Concavity

- **Concave Up (C. U.)**



- bent so that the graph opens upward
- $f''(x)$ is positive over this interval
- tangents are below the graph

- **Concave Down (C. D.)**

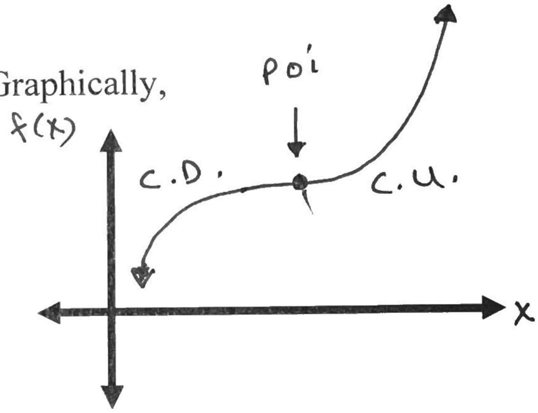


- bent so that the graph opens downward
- $f''(x)$ is negative over this interval
- tangents are above the graph

Point of Inflection (P. O. I.)

The point where the graph changes from concave up to concave down or vice versa is called a P. O. I.

Graphically,



Note: $f''(x) = 0$ at a P. O. I.

Test for Concavity

- If $f''(x) > 0$ for all x on some interval, then the graph of $f(x)$ is concave up on that interval.

- If $f''(x) < 0$ for all x on some interval, then the graph of $f(x)$ is concave down on that interval.

To Find Possible P. O. I.

Solve $f''(x) = 0$ or determine where $f''(x)$ DNE. Use an interval chart to determine on which intervals $f''(x)$ is positive or negative.

Example 1: Determine where the following curves are C. U. or C. D. and locate any P. O. I.

a) $f(x) = x^3 - 3x^2 + 4x$

$f'(x) = 3x^2 - 6x + 4$

$f''(x) = 6x - 6$

$= 6(x-1)$

possible p.o.i.

$f''(x) = 0$

$6(x-1) = 0$

$\therefore x = 1$

Interval	$(-\infty, 1)$	$(1, \infty)$
$x-1$	$-$	$+$
$f''(x)$	$-$	$+$
$f(x)$	C.D.	C.U.

$\therefore$ poi at $(1, 2)$

concave down @ $(-\infty, 1)$

concave up @ $(1, \infty)$

b) $f(x) = \sqrt[3]{x}$

$$f'(x) = \frac{1}{3} x^{-\frac{2}{3}}$$

$$f''(x) = \frac{-2}{9} x^{-\frac{5}{3}} = \frac{-2}{9\sqrt[3]{x^5}}$$

possible poi $f''(x)$ DNE @ $x = 0$

$$\frac{-2}{9x^{5/3}} \Rightarrow x = 0$$

Interval	$(-\infty, 0)$	$(0, \infty)$
$-\frac{2}{9}$	$-$	$-$
$x^{5/3}$	$-$	$+$
$f''(x)$	$+$	$-$
$f(x)$	C.U.	C.D.

$\therefore$ poi @ $(0,0)$

concave up @ $(-\infty, 0)$

concave down @ $(0, \infty)$

Cusps and Vertical Tangents

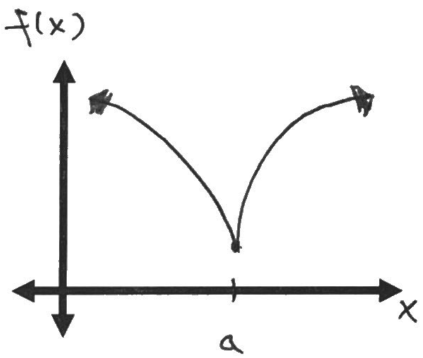
- A cusp is a pointy bend on the graph where no unique tangent line exists.

- A cusp is a good example of the situation where a function can have a maximum or a minimum at a point where the derivative of the function DNE.

- To locate a cusp, determine where $f'(x)$ DNE and then test the concavity on either side of that critical number.

  → If the concavity remains the same there is a **cusp**.

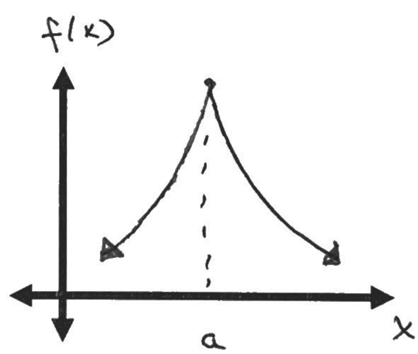
  → If the concavity changes there is a **vertical tangent**.



* cusp at x = a

* NO p.o.i.

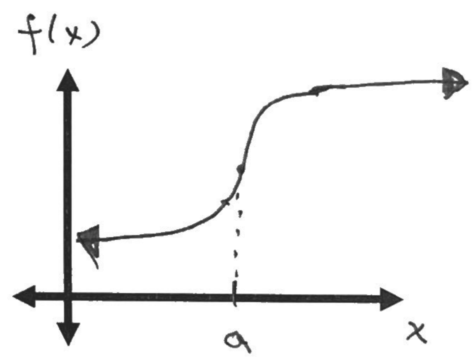
* local minimum



* cusp at x = a

* NO p.o.i.

* local maximum



* vertical tangent at x = a.

* p.o.i. at the vertical tangent.

Example 2: Verify that $f(x) = x^{\frac{2}{3}}$ has a cusp at $x = 0$.

- $f'(x) = \frac{2}{3} x^{-\frac{1}{3}} = \frac{2}{3\sqrt[3]{x}}$

  critical #: $f'(x)$ DNE @ $x = 0$

Interval	$(-\infty, 0)$	$(0, \infty)$
$x^{\frac{1}{3}}$	$-$	$+$
$f'(x)$	$-$	$+$
$f(x)$	$\downarrow$	$\uparrow$

  local min @ $(0,0)$

- $f''(x) = -\frac{2}{9} x^{-\frac{4}{3}}$

  possible poi

  $f''(x)$ DNE @ $x = 0$

Interval	$(-\infty, 0)$	$(0, \infty)$
$-\frac{2}{9}$	$-$	$-$
$x^{-\frac{4}{3}}$	$+$	$+$
$f''(x)$	$-$	$-$
$f(x)$	C.D.	C.D.

  No change in concavity

- $\Rightarrow$ cusp @ $(0,0)$

Test for Local Extrema using the 2^{nd} Derivative Test

Given c is a critical number, then

→ If $f'(c)=0$ and $f''(c)>0$, then $f(x)$ has a local minimum at c.

→ If $f'(c)=0$ and $f''(c)<0$, then $f(x)$ has a local maximum at c.

→ If $f'(c)=0$ and $f''(c)=0$, then the 2^{nd} Derivative Test fails.

Example 3: Use the 2^{nd} Derivative Test to find the local extrema of $f(x)=3x^5-5x^3$.

$= x^3(3x^2-5)$

$f'(x) = 15x^4 - 15x^2$

$= 15x^2(x^2-1)$

critical #'s: $x = 0, \pm 1$

$f''(x) = 60x^3 - 30x$

$f''(0) = 0$

Test fails.

* use 1^{st} derivative test!

$f''(1) = 30 > 0$

$\Rightarrow$ local min.

@ $(1,-2)$

$f''(-1) = -30 < 0$

$\Rightarrow$ local max.

@ $(-1,2)$